Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/21/2007
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Project Title:	HZE Radiation Effects on Neuroinflammation: Role of COX-2		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14642-0001	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	06/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed back to 06/01/2008 per S. Krenek/JSC (8/07) NOTE: end date revised to 6/30/2008 per JSC info (10/06) originally in task book as Michael O'Banion (changed 2/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olschowka, John (University of Rochester Medical Center) Williams, Jacqueline (University of Rochester Medical Center) Hurley, Sean (University of Rochester Medical Center)		
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Task Description:	Localized activation of glial cells and upregulation of inflammatory mediators, a process called neuroinflammation, is a common feature of brain injury that may contribute to neuropathological processes in diseases ranging from stroke to Alzheimer's. Our laboratory has investigated acute and chronic neuroinflammatory changes following gamma irradiation of the brain in mice. We find dose dependent changes in many indices of neuroinflammation including morphological activation of astrocytes and microglia, induction of pro-inflammatory cytokines and chemokines, upregulation of prostaglandin production, disruption of blood-brain barrier and edema over a time course of hours to days following irradiation. Similar changes are also observed months after injury in association with white matter damage, modeling the clinical entity known as radiation necrosis. Our efforts have specifically focused on the role of prostaglandins in brain neuroinflammation following radiation injury and have revealed that pharmacological inhibition of cyclooxgygenase-2, an inducible isoform first described in our laboratory, attenuates many CNS responses including induction of cytokines and tissue edema. We plan to establish whether similar neuroinflammatory changes occur in mouse brain following exposure to the unique damage imposed by HZE particles. There is a degree of uncertainty in our understanding of the standard radiobiological parameters (e.g. RBE) with respect to the impact of this type of radiation in the CNS. Therefore, in our first aim we will endeavor to use a sufficient dose range in order to compare outcomes with the physiologic and pathologic endpoints established in our current data using gamma radiation. In our second aim, we will use specific inhibitors of cyclooxygenase-2 to determine if prostaglandins play a key role in the neuroinflammatory response to HZE particles. These studies address important fundamental questions about the interaction of HZE radiation with brain tissue and may lead to simple treatment options that provide protection from neuroinflammation and its potential contribution to neurological problems arising in future space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Neuroinflammation is a reaction of brain to any injurious process whether it be an acute response to radiation or long term response to neurodegenerative disease. There is clear evidence that this reatcion can contribute to brain disease. Our study includes experiments aimed at curbing neuroinflammation. Successful demonstration of this effect has implications for treatement of brain disease where neuroinflammation is a major component.
Task Progress:	We have continued to collect data from a long-term study of mice that were initially irradiated with 6 Gy 1000 MeV/amu 56Fe radiation in October, 2005. Although originally scheduled to only last six months, we carried the study out to 1 year with additional mice irradiated for this contingency. All mice have now been sacrificed and their tissue processed. Molecular characterization that includes real-time quantitative RT-PCR measures of mRNA for COX-2, the proinflammatory cytokines, IL-1ß and TNFa, the chemokine MCP-1, as well as cell activation markers for astrocytes (GFAP), microglia (MHC-II), and endothelial cells (ICAM-1) shows modest activation at some, but not the majority of time points analyzed. Unfortunately, with the many samples represented, the data was obtained from several different analyses, making absolute comparisons across time difficult. Therefore, these RNA measures will be repeated to achieve the very best quality data. Nevertheless, it is apparent that HZE radiation at a dose of 6 Gy can have a significant dampening effect on glial activation in the first week following exposure. Moreover, these results confirm our findings in the first pilot study that neuroinflammatory marker expression following HZE irradiation does not seem to mirror expression patterns with gamma radiation. These findings prompted us to modify the proposed dose-response experiment in Aim 1 to focus on a full dose response over an acute time course with a direct comparison to gamma irradiation. This study is described in more detail below. In addition to the molecular findings in the time course study with 6 Gy HZE exposure, we analyzed glial activation in histological sections. Although these analyses are not complete, we do see evidence for increased glial activation by six months. Interestingly, the profiles of MHC-II positive cells in brain 6 months after 6 Gy HZE radiation are consistent with infiltrating macrophages. Thus, there are long-term morphologic responses in brain following HZE irradiation. Whether such changes are associated with neuronal damage or might be seen with lower doses that are more relevant to space travel is unknown. The findings above have led to a new hypothesis, namely that the neuroinflammatory response following HZE particle exposure is fundamentally different from that following low LET radiation. To explore this idea, we conducted a full dose response study using 1000 MeV/amu 56Fe and a parallel set of gamma radiation doses with tissues collected at a very early time point (4 h) for molecular analyses and at 3 days for histological evidence of glial activation and possible cell infiltration. The study was carried out during the October, 2006 NRSL run for HZE irradiation and using a source at Rochester for the low LET exposure. Initial analysis of mRNA expression at the acute time point clearly reveals a unique “inverted U” shaped response for a number of target genes with HZE radiation. These results confirm our hypothesis about differences existing between the effects of HZE and low LET radiation, at least at the time point examined. We have followed up on our observation of evidence for in vivo dsDNA breaks and axonal damage in brain by exposing a small number of mice to whole body irradiation with 1 Gy 1000 MeV/amu 56Fe and collecting tissues, including brain, liver, spleen and testis from these animals at 2 h. This time was chosen on the basis of when we expected to see apoptosis of injured cells. Indeed, TUNEL staining revealed clear evidence of apoptosis in several of the tissues examined, but not in brain. Though other time points should be examined, these findings are consistent with observations from other laboratories that HZE irradiation does not have a profound effect on neuron number, which might be expected since these cells are non-dividing. Given the dramatic evidence of DNA damage recorded with our other method, this suggests that neuronal cells (neurons and glia) are particularly resistant to the DNA damaging effects of HZE particles.
Bibliography Type:	Description: (Last Updated: 01/07/2026)
Abstracts for Journals and Proceedings	Hurley SD, Williams JP, Olschowka JA, Vazquez M, Trojanczyk L, O’Banion MK. "Acute measures of neuropathology following HZE brain irradiation." Presented at 4th International Workshop on Space Radiation Research, Moscow – St. Petersburg, Russia, June 05-09, 2006. 4th International Workshop on Space Radiation Research, Proceedings: 87P. June 2006. , Jun-2006

Abstracts for Journals and Proceedings	O'Banion MK, Williams JP, Olschowka JA, Vazquez M, Trojanczyk L, Hurley SD. "HZE radiation induces neuroinflammatory changes in mouse brain. " Presented at 4th International Workshop on Space Radiation Research, Moscow – St. Petersburg, Russia, June 05-09, 2006. 4th International Workshop on Space Radiation Research, Proceedings: 50P. June 2006. , Jun-2006
Abstracts for Journals and Proceedings	O'Banion MK, Hurley SD, Olschowka JA, Williams JP. "HZE radiation induces neuroinflammation in mouse brain. " Presented at 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5-8, 2006. 53rd Annual Meeting of the Radiation Research Society, Proceedings: 49P. November 2006. , Nov-2006
Abstracts for Journals and Proceedings	O'Banion MK, Williams JP, Olschowka JA, Vazquez M, Trojanczyk L, Hurley SD. "HZE radiation induces neuroinflammation in mouse brain. HZE radiation effects on neuroinflammation: role of COX-2." Presented at NASA HRP Investigator's Workshop, League City, TX, February 11-14, 2007. http://www.dsls.usra.edu/meetings/hrp2007/pdf/Radiation2/Rad2.pdf , Feb-2007